

Ali Hatem Tahssin

Computer and AI Engineering Student

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Education

Ain Shams University (iCHEP)

Computer and Artificial Intelligence Engineering | Sep 2023 – Present | Graduation: 2027

Cairo, Egypt

University of East London (Dual Degree)

Computer and Artificial Intelligence Engineering | Sep 2023 – Present | Graduation: 2027

Cairo, Egypt

Training & Certification

Digital Twin Technologies by Siemens

April 2026

- Covered the 5D digital twin lifecycle, virtual product design and validation, and smart system architectures using the Innexis Virtual System Interconnect framework
- Experienced hands-on with ROS, FMI/FMUs, sensor and scenario modeling, and Siemens simulators; capstone: designed, simulated, and deployed a digital twin on AWS VPC

Data Science Training by CLS Learning

Nov 2025

- Covered Python, SQL, statistics and probability, and linear algebra; trained supervised and unsupervised models
- Dived into deep learning, NLP, Gen AI, and MLOps
- Built data pipelines with NumPy, Pandas, and scikit-learn including preprocessing, feature engineering, SMOTE for class imbalance, and Stratified K-Fold cross-validation

Linux Administration by Edges for Training

Aug 2025

- Mastered Linux CLI, advanced Bash scripting, user/group permission management, process and package management, networking, services and systemd, disks and filesystems, and Makefile/CMake build systems; applied skills across hands-on labs and a final capstone project

Competitive Programming Summer Training by Coach Academy

Aug 2024

- Intensive training in algorithm design, complexity analysis, STL containers, dynamic programming, and recursion; competed in an on-site programming contest

Projects

Async Health Monitor System | *Python*

- Built an async health-monitoring backend in FastAPI with per-service check loops, PostgreSQL storage via SQLAlchemy, and status-change alerts
- Containerized the app with Docker and Compose for local dev and production, using an override file to avoid touching the prod config
- Provisioned AWS EC2 infrastructure with Terraform, including networking and automated Docker install on boot.
- Wrote a 4-layer test suite (unit, integration, e2e, Postman) with a 4-stage GitHub Actions CI/CD pipeline: test, build, push, deploy.

Distributed LLM Inference System | *Python*

- Architected a fault-tolerant master-worker system on AWS EC2, using NGINX to load-balance requests into a FastAPI scheduler dispatching inference across 3 worker nodes
- Built 3 load-balancing strategies (Round Robin, Least Connections, CPU aware), zero-downtime hot-swapping, heartbeat failover excluding unresponsive workers, and per-worker semaphores to prevent OOM under load
- Engineered a hybrid RAG pipeline (ChromaDB and keyword fallback) feeding a built in Ollama LLM
- Load-tested across 8 waves scaling 10 to 100 concurrent users, hitting 100% success rate at sub-8.2s P95 latency; built a live dashboard for per-worker CPU/memory/connection metrics with 2s refresh

Deployed Reddit Website Clone | *JavaScript / MERN*

- Developed a full-stack social media website simulating Reddit using Node/Express, MongoDB, and React; containerizing each layer
- Deployed with JWT authentication, user karma system, community system, nested comments, upvote/downvote system, paginated feed, full-text search, image post support, and generative AI post summarization via REST API; layered MVC architecture

Agent Tracker Database Management System | *C# / MongoDB*

- Designed MongoDB schema with JSON Schema validation and compound indexes; built 4 aggregation pipelines (SLA breach detection, agent workload, case history); developed WinForms CRUD app via .NET MongoDB driver

Technical Skills

- Cloud & DevOps:** AWS, Linux, Docker, Docker Compose, CI/CD, GitHub Actions, Terraform, Nginx, VMs, SSH, Bash Scripting
- Backend & APIs:** Python, FastAPI, Node.js, Express, REST API, JWT, async/await, SQLAlchemy, Mongoose, Pydantic
- Databases:** PostgreSQL, MongoDB, SQL, Redis
- Programming Languages:** Python, JavaScript, TypeScript, Java, C++, C#
- Testing & Tools:** Pytest, Postman, Git/GitHub
- Frontend:** React, HTML, CSS
- Languages:** Arabic (native), English (fluent)
- Soft Skills:** Research, Problem Solving, Communication, Active Listening, Time Management, Teamwork